

Deployment of a Website Using Kubernetes

This project demonstrates how a static website can be deployed and managed using Kubernetes. Kubernetes is an open-source container orchestration platform that automates the deployment, scaling, and management of containerized applications. In this project, a website is packaged into a Docker container and deployed to a Kubernetes cluster using Deployment and Service resources. The application can be accessed through a web browser, and Kubernetes ensures high availability, scalability, and reliability.

Objectives

- To understand Kubernetes architecture and workflow.
- To deploy a website using Kubernetes.
- To manage application replicas using Deployments.
- To expose the application using Kubernetes Services.
- To learn container orchestration concepts.

System Architecture

The deployment process follows these steps:

1. Develop the website.
2. Create a Docker image.
3. Push the Docker image to Docker Hub (optional).
4. Create Kubernetes Deployment YAML.
5. Create Kubernetes Service YAML.
6. Apply YAML files using kubectl.
7. Kubernetes creates Pods.
8. Users access the website through the Service.

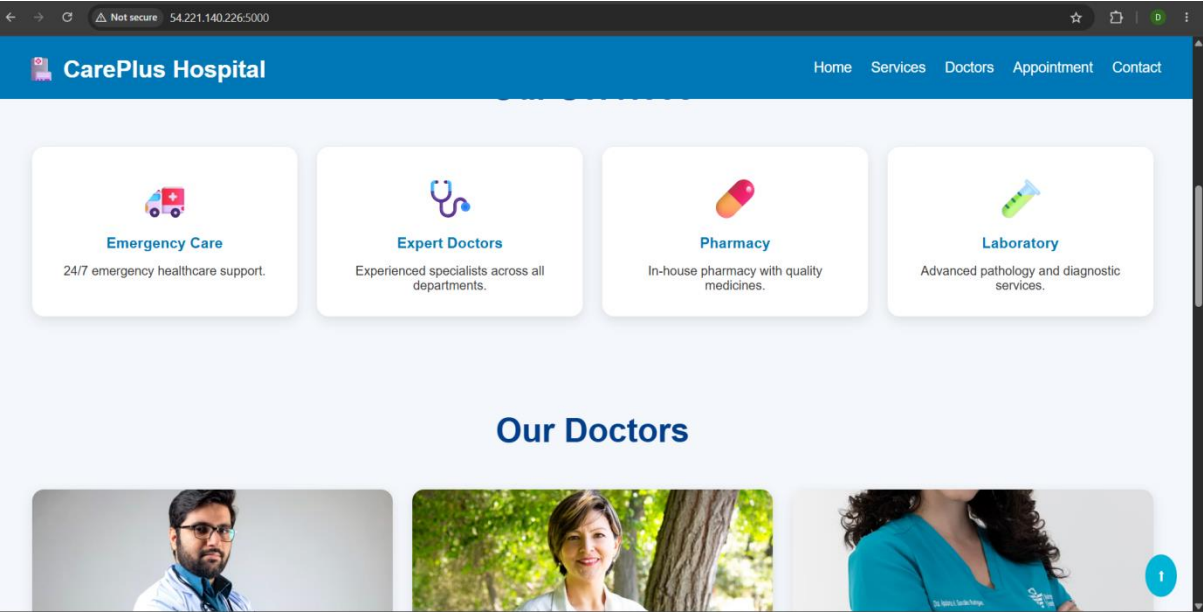
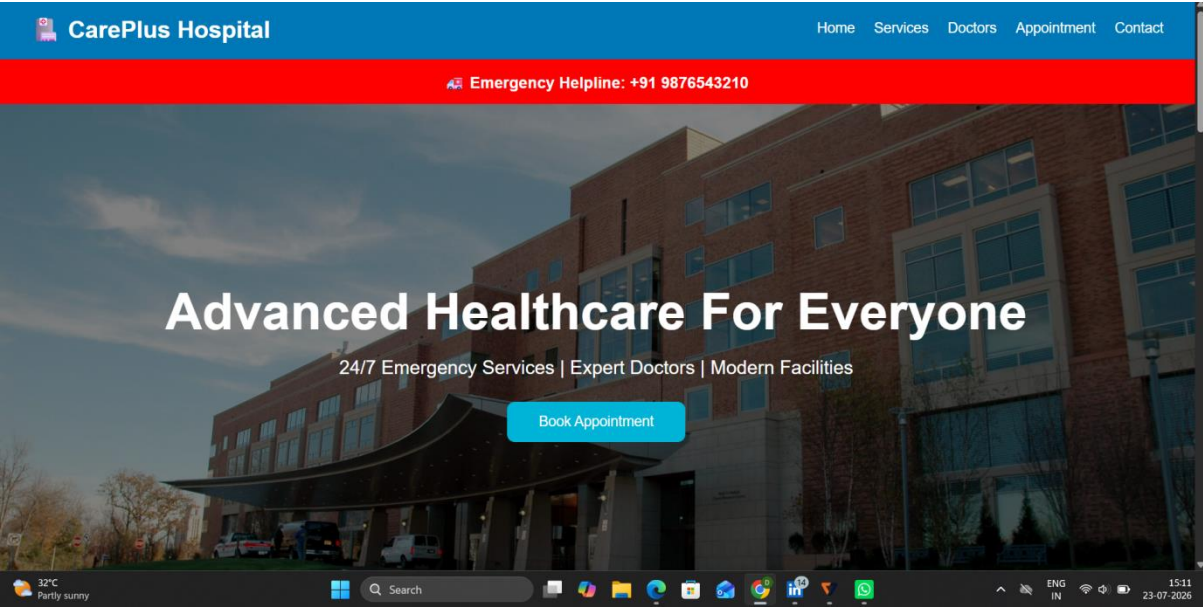
Features of Kubernetes

- Automatic deployment
- Self-healing
- Auto scaling
- Load balancing
- High availability
- Rolling updates
- Rollback support
- Service discovery
- Resource management
-

```
handling connection for 5000
handling connection for 5000
Croot@ip-172-31-19-175:/home/ubuntu# history
1  chmod +x kubect1
2  mv kubect1 /usr/local/bin/
3  kubect1 version --client
4  curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
5  install minikube-linux-amd64 /usr/local/bin/minikube
6  minikube version
7  cd /home/ubuntu
8  minikube start --driver=docker --force
9  kubect1 cluster-info
10 kubect1 get nodes
11 touch Dockerfile pod.yml app.py
12 cat > app.py
13 cat > Dockerfile
14 cat > pod.yml
15 docker build -t ubuntu-hospital-web .
16 kubect1 apply -f pod.yml
17 cat pod.yml
18 ls
19 cat > pod.yml
20 kubect1 apply -f pod.yml
21 kubect1 get pods
22 kubect1 describe hospital-pod
23 kubect1 describe pod hospital-pod
24 kubect1 port-forward hospital-pod 5000:5000 --address=0.0.0.0
25 history
oot@ip-172-31-19-175:/home/ubuntu#
```

i-062817b2963e48d35 (divi)

PublicIPs: 54.221.140.226 PrivateIPs: 172.31.19.175



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Advantages

- Easy application deployment
- Supports scalability
- High availability
- Fault tolerance
- Automatic recovery
- Efficient resource utilization
- Simplified management
- Cloud portability

Future Enhancements

- Configure Ingress for domain-based routing.
- Enable HTTPS using SSL/TLS.
- Add Horizontal Pod Autoscaler.
- Implement monitoring with Prometheus and Grafana.
- Configure CI/CD using GitHub Actions or Jenkins.
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Conclusion

This project successfully demonstrates the deployment of a website using Kubernetes. The website was containerized with Docker and deployed to a Kubernetes cluster using Deployments and Services. Kubernetes efficiently manages application availability, scalability, and fault tolerance. This project provides practical knowledge of modern container orchestration and cloud-native application deployment.